Extracting AC resistance from SIwave S-parameters

Detailed mathematics, model selection and validation

For SIwave 2024 R2 SYZ exports and Q3D comparisons

Can this be done? Yes. Frequency-dependent series resistance can be extracted from S-parameters when the physical network supports an identified transmission-line or lumped-circuit model. The established distributed method is

$$\mathbf{S}(f) \longrightarrow \mathbf{T}(f) \longrightarrow \gamma(f), Z_c(f) \longrightarrow \boxed{R_{\mathrm{AC}}(f) = \ell\,\mathrm{Re}\{\gamma(f) Z_c(f)\}}.$$

For an arbitrary board, a two-port file alone does not uniquely identify the copper resistance of each internal conductor.

What this guide establishes. It derives the network conversions, the distributed RLGC extraction, the short-line and lumped limits, and the power interpretation. It explains exactly when each result is a physical conductor-path resistance and when it is only a terminal equivalent.

Evidence and scope. The mathematics is supported by the research listed in Section 12. A fresh numerical test generates S-parameters from a known passive uniform-line model and recovers its series resistance. This is a circuit-model validation, not a SIwave/Q3D field solve. The user's actual script, S2P data, Q3D CSV and port definitions have not been inspected.

Interpretation of the reported discrepancy. A script result of 0.5–1% *of* Q3D means a resistance about 100–200 times smaller. No particular cause is assumed here.

Reading route

Sections 1–5 give the main derivation and branch/DC treatment. Sections 6–8 connect lumped RF methods to physical loss and Q3D. Sections 9–11 give validation, practical processing and implementation. Section 12 contains clickable research links and source notes.

Prepared 14 October 2026. Equations are typeset with LaTeX. Reusable LaTeX expressions are included in Section 11.

1 Define the quantity before extracting it

The desired quantity is normally a *series resistance* associated with a stated current path. It must be distinguished from the real part of an arbitrary input impedance, which depends on how the other port is terminated.

Use the $e^{j\omega t}$ convention, $\omega = 2\pi f$, and RMS phasors. Currents I_1, I_2 enter the two-port at their respective positive terminals. Let V_1, V_2 be the corresponding port voltages. Then

$$\mathbf{V} = \mathbf{ZI}, \qquad \mathbf{I} = \mathbf{YV}, \qquad \mathbf{Z} = \mathbf{Y}^{-1} \quad \text{when invertible.} \tag{1}$$

These are terminal relationships; they do not specify an internal circuit topology.

Power and the meaning of a resistance

The net real power entering the network is

$$P = \mathrm{Re}\{\mathbf{I}^H\mathbf{V}\} = \mathbf{I}^H \frac{\mathbf{Z} + \mathbf{Z}^H}{2}\mathbf{I}. \tag{2}$$

The superscript H denotes conjugate transpose. For a reciprocal network, $\mathbf{Z}^T = \mathbf{Z}$, so its Hermitian real part is the elementwise real matrix. For a general network, use the Hermitian expression above.

If port 2 is open, $I_2 = 0$, then $P = |I_1|^2 \,\mathrm{Re}(Z_{11})$. If the intended branch current is I and its own dissipated power is P_{cond}, the conductor resistance is

$$\boxed{R_{\mathrm{cond}} = \frac{P_{\mathrm{cond}}}{|I|^2}}. \tag{3}$$

An S-parameter file determines total terminal behavior. Without a model or additional field information, it does not partition power among copper, dielectric, components and radiation. Peak phasors introduce a factor 1/2 in both power and its current normalization; consistent conventions give the same resistance.

Distributed quantities and units

For a uniform two-conductor line, use primes for per-unit-length parameters:

Symbol	Meaning	Unit
R', L'	Longitudinal series resistance and inductance	Ω/m, H/m
G', C'	Transverse shunt conductance and capacitance	S/m, F/m
$\gamma = \alpha + j\beta$	Propagation constant	1/m
Z_c	Characteristic impedance of the line	Ω
Z_0	Reference impedance used to normalize S	Ω
ℓ	Physical length of the modeled line	m

Z_0 and Z_c are different. A 50-Ω S-parameter export does not mean the conductor path itself has 50-Ω characteristic impedance.

2 From complex S-parameters to ABCD

Assume ordinary single-ended ports with the *same positive real* reference impedance Z_0. Define incident and reflected wave amplitudes by

$$a_k = \frac{V_k + Z_0 I_k}{2\sqrt{Z_0}}, \qquad b_k = \frac{V_k - Z_0 I_k}{2\sqrt{Z_0}}, \qquad \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \mathbf{S} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}. \tag{4}$$

Thus $V_k = \sqrt{Z_0}(a_k + b_k)$ and $I_k = (a_k - b_k)/\sqrt{Z_0}$. From $b_2 = S_{21}a_1 + S_{22}a_2$,

$$a_1 = \frac{b_2 - S_{22}a_2}{S_{21}}, \qquad b_1 = \frac{S_{11}}{S_{21}} b_2 + \left(S_{12} - \frac{S_{11}S_{22}}{S_{21}} \right) a_2. \tag{5}$$

Insert $a_2 = (V_2 + Z_0 I_2)/(2\sqrt{Z_0})$ and $b_2 = (V_2 - Z_0 I_2)/(2\sqrt{Z_0})$ into V_1, I_1. Collect coefficients using the chain convention

$$\boxed{\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}.} \tag{6}$$

The minus sign makes the output chain current point in the forward direction, whereas I_2 enters the network. The resulting coefficients are

$$A = \frac{(1 + S_{11})(1 - S_{22}) + S_{12}S_{21}}{2S_{21}}, \tag{7}$$

$$B = Z_0 \frac{(1 + S_{11})(1 + S_{22}) - S_{12}S_{21}}{2S_{21}}, \tag{8}$$

$$C = \frac{(1 - S_{11})(1 - S_{22}) - S_{12}S_{21}}{2Z_0 S_{21}}, \tag{9}$$

$$D = \frac{(1 - S_{11})(1 + S_{22}) + S_{12}S_{21}}{2S_{21}}. \tag{10}$$

Here A, D are dimensionless, B is in ohms and C is in siemens. For a reciprocal two-port,

$$AD - BC = \frac{S_{12}}{S_{21}} = 1. \tag{11}$$

All operations use **complex** S-parameters, including their phase. Taking magnitudes first destroys information needed to separate resistance and reactance. These formulas also become numerically problematic when S_{21} is extremely small. Unequal or complex port references require a generalized wave conversion or explicit renormalization.

3 Derive the transmission-line model

Let x run along a uniform line from its input to its output. Let $I(x)$ point toward increasing x, and let $V(x)$ be voltage between the conductor and its specified return. Define

$$z = R' + j\omega L', \qquad y = G' + j\omega C'. \tag{12}$$

An infinitesimal section dx has a series drop $zI\,dx$ and a transverse leakage current $yV\,dx$. Kirchhoff's laws give the telegrapher equations:

$$\frac{dV}{dx} = -zI, \qquad \frac{dI}{dx} = -yV. \tag{13}$$

Differentiate the first equation and substitute the second:

$$\frac{d^2V}{dx^2} = zyV = \gamma^2 V, \qquad \gamma^2 = zy. \tag{14}$$

Consequently,

$$V(x) = V^+ e^{-\gamma x} + V^- e^{\gamma x}, \tag{15}$$

$$I(x) = \frac{1}{Z_c}\left(V^+ e^{-\gamma x} - V^- e^{\gamma x}\right), \qquad Z_c = \frac{z}{\gamma}. \tag{16}$$

The first term is the forward traveling wave; the second is the backward wave. Their relationship requires

$$\boxed{\gamma Z_c = z, \qquad \frac{\gamma}{Z_c} = y.} \tag{17}$$

Evaluate V and I at $x = 0$ and $x = \ell$. Eliminating the wave amplitudes yields

$$V(0) = \cosh(\gamma\ell)V(\ell) + Z_c \sinh(\gamma\ell)I(\ell), \tag{18}$$

$$I(0) = \frac{\sinh(\gamma\ell)}{Z_c}V(\ell) + \cosh(\gamma\ell)I(\ell). \tag{19}$$

Since $I(\ell) = -I_2$ at the terminal, the ABCD representation becomes

$$\boxed{\mathbf{T} = \begin{bmatrix} \cosh\theta & Z_c \sinh\theta \\ \sinh\theta / Z_c & \cosh\theta \end{bmatrix}, \qquad \theta = \gamma\ell.} \tag{20}$$

This establishes the inverse problem: obtain θ and Z_c from terminal data, then recover z and y. The model assumes one relevant propagation mode, uniform longitudinal parameters, and reference planes at the ends of that uniform section. It does not automatically describe an entire board with branches, connectors, pads, vias or changing cross-sections.

4 Extract the series resistance and the other RLGC terms

For an ideal uniform reciprocal line, $A = D = \cosh\theta$. Therefore, using a physically chosen branch,

$$\theta = \operatorname{acosh}\left(\frac{A+D}{2}\right), \qquad \gamma = \frac{\theta}{\ell}, \qquad Z_c = \frac{B}{\sinh\theta}. \tag{21}$$

Averaging A and D helps only with small numerical asymmetry. It cannot turn a nonuniform structure into a uniform line. When valid, multiplying the expressions gives

$$\gamma Z_c = \frac{1}{\ell}\frac{\theta B}{\sinh\theta}. \tag{22}$$

The complete extraction is then

$$R' = \frac{1}{\ell}\operatorname{Re}\left(\frac{\theta B}{\sinh\theta}\right), \qquad L' = \frac{1}{\omega\ell}\operatorname{Im}\left(\frac{\theta B}{\sinh\theta}\right), \tag{23}$$

$$G' = \frac{1}{\ell}\operatorname{Re}\left(\frac{\theta C}{\sinh\theta}\right), \qquad C' = \frac{1}{\omega\ell}\operatorname{Im}\left(\frac{\theta C}{\sinh\theta}\right). \tag{24}$$

Both L' and C' expressions require $\omega \neq 0$. The total longitudinal series resistance is

$$\boxed{R_{\mathrm{AC}}(f) = \ell R'(f) = \operatorname{Re}\left[\frac{\theta(f)}{\sinh\theta(f)}B(f)\right].} \tag{25}$$

This is the central S-to-ACR formula under the uniform-line model. The length cancels for total resistance but is needed for per-unit-length quantities and physical branch interpretation.

An alternative is $Z_c = \sqrt{B/C}$. Its square-root sign must be chosen consistently with the branch of θ; independent principal square roots can give the wrong sign. Using $B/\sinh\theta$ explicitly ties them together.

What R' represents physically

In this two-conductor mode, R' is the longitudinal series loss for the conductor *and its specified return*, under the mode's current distribution. It includes modeled conductor loss mechanisms such as skin/proximity redistribution when those mechanisms are captured by the solver. It is not automatically the resistance of the positive conductor alone.

Separating R' and G' is meaningful because the assumed model separates series and shunt mechanisms. For a network that does not fit that model, the extracted terms may be frequency-dependent terminal equivalents without that physical interpretation.

The S-to-RLGC route is described in the research literature [1,2] and implemented in MathWorks' `s2rlgc` [5]. That establishes feasibility; it does not certify an arbitrary SIwave export as a suitable uniform line.

5 Branch selection, singular points and the DC limit

Why the inverse is not single-valued

If $x = \cosh\theta$, then every

$$\theta_{\sigma,k} = \sigma\,\mathrm{acosh}(x) + j2\pi k, \qquad \sigma \in \{-1,+1\}, \quad k \in \mathbb{Z} \tag{26}$$

produces the same x. A single frequency point cannot identify the accumulated propagation phase. Changing k changes $\theta B/\sinh\theta$ and can therefore change the extracted resistance, not only the inductance.

For a passive forward wave choose $\alpha = \mathrm{Re}(\gamma) \geq 0$, then track phase continuously across an adequately sampled frequency sweep. Use a known low-frequency seed or a delay/geometry estimate to select the initial phase branch. Candidate selection at subsequent points should consider the previous solution, expected delay and continuity of the extracted parameters. Phase jumps between samples must be small enough to avoid aliasing.

Continuity and nonnegative loss are necessary checks, not a proof of uniqueness. A sparse sweep or an unknown electrically long section may require a second length, a delay estimate, de-embedding or a constrained broadband fit. The extraction literature discusses branch discontinuities and fixture effects [1,2].

Stable evaluation for a short electrical length

Direct evaluation is vulnerable when $\theta \to 0$ because $\sinh\theta$ is also small. The removable singularity has the series

$$F(\theta) = \frac{\theta}{\sinh\theta} = 1 - \frac{\theta^2}{6} + \frac{7\theta^4}{360} - \frac{31\theta^6}{15120} + \cdots . \tag{27}$$

Use the series for sufficiently small $|\theta|$; its error can be bounded by the neglected terms. At $\theta = j\pi n$ with nonzero integer n, both ideal B and C vanish and the extraction is ill-conditioned. The matrix can then be $\pm\mathbf{I}$ regardless of Z_c. Use neighboring frequencies or fit a broadband model.

Exactly DC

For an insulated line, $y \to 0$, so $\theta \to 0$ and the characteristic-impedance representation can diverge even though the terminal matrix remains finite:

$$\mathbf{T}(0) = \begin{bmatrix} 1 & \ell R'_{\mathrm{DC}} \\ 0 & 1 \end{bmatrix}, \qquad R_{\mathrm{DC}} = \mathrm{Re}\{B(0)\}. \tag{28}$$

This assumes negligible DC shunt leakage. If DC leakage is appreciable, retain the appropriate DC network model. A minimum nonzero AC frequency is not an actual DC solution; report it as such. A limiting extrapolation also requires an explicitly validated model.

6 Lumped extraction and RF series/shunt methods

For an electrically short path, the series expansion of Section 5 gives

$$R_{\mathrm{AC}} \approx \mathrm{Re}(B), \qquad B \approx z\ell. \tag{29}$$

Electrical shortness is measured by $|\gamma\ell|$, not by a universal frequency threshold. A low-resistance path with parasitic shunts still needs an appropriate circuit representation.

Exact pi-network interpretation

Suppose the physical model has an input shunt y_1, series arm z_s and output shunt y_2, all referenced to the same return. Its chain matrix factors as

$$\mathbf{T} = \begin{bmatrix} 1 & 0 \\ y_1 & 1 \end{bmatrix} \begin{bmatrix} 1 & z_s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ y_2 & 1 \end{bmatrix} = \begin{bmatrix} 1 + z_s y_2 & z_s \\ y_1 + y_2 + y_1 z_s y_2 & 1 + y_1 z_s \end{bmatrix}. \tag{30}$$

Thus $z_s = B$, $y_1 = (D-1)/B$ and $y_2 = (A-1)/B$, where defined. Its nodal equations also give

$$\mathbf{Y} = \begin{bmatrix} y_1 + 1/z_s & -1/z_s \\ -1/z_s & y_2 + 1/z_s \end{bmatrix}, \qquad \boxed{z_s = -\frac{1}{Y_{12}}.} \tag{31}$$

A reciprocal two-port with nonzero B can be represented by such a terminal pi equivalent. Its equivalent arm is not necessarily the physical conductor resistance. For a uniform distributed line, $B = Z_c \sinh\theta$, whereas its integrated series impedance is $z\ell = \theta Z_c$. Their difference is exactly the correction factor $F(\theta)$.

Series-through and shunt-through fixtures

For a series-only impedance z_s, $A = D = 1$, $B = z_s$, $C = 0$. The forward transmission is

$$S_{21} = \frac{2}{2 + z_s/Z_0}, \qquad \boxed{z_s = 2Z_0 \frac{1 - S_{21}}{S_{21}}.} \tag{32}$$

For a shunt-only impedance z_p, $A = D = 1$, $B = 0$, $C = 1/z_p$. Hence

$$S_{21} = \frac{2}{2 + Z_0/z_p}, \qquad \boxed{z_p = \frac{Z_0}{2} \frac{S_{21}}{1 - S_{21}}.} \tag{33}$$

These are different DUT connections. Take the real part after the complex calculation. Keysight explicitly documents both methods [6]; neither S_{21}-only formula is a general extractor for an arbitrary layout.

7 Terminal resistance, loading and power

To interpret an existing script, identify the quantity its formula computes. From $V_2 = Z_{21}I_1 + Z_{22}I_2$ and a load across port 2, $V_2 = -Z_L I_2$, one obtains

$$I_2 = -\frac{Z_{21}}{Z_{22} + Z_L} I_1, \qquad Z_{\text{in}} = Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22} + Z_L}. \tag{34}$$

Open, short and matched loads therefore describe different excitations:

Condition	Input impedance
Port 2 open: $I_2 = 0$	Z_{11}
Port 2 short: $V_2 = 0$	$Z_{11} - Z_{12}Z_{21}/Z_{22} = 1/Y_{11}$
Port 2 matched to Z_0	$Z_0(1 + S_{11})/(1 - S_{11})$

The matched input's real part includes power delivered to the external load as well as network loss. It cannot generally be equated to the network's internal dissipative resistance.

An equal-and-opposite-current measurement

For $I_1 = I$, $I_2 = -I$ and $V_d = V_1 - V_2$,

$$Z_d = \frac{V_d}{I} = Z_{11} + Z_{22} - Z_{12} - Z_{21}, \qquad P = |I|^2 \operatorname{Re}(Z_d). \tag{35}$$

This can be useful if the actual terminal arrangement corresponds to the desired path. It is not automatically a through-line copper resistance at high frequency, when shunt currents change the internal current distribution. No additional factor of two or four is needed under these definitions.

Correct complex arithmetic

If $Y_{11} = g + jb$, then

$$\operatorname{Re}\left(\frac{1}{Y_{11}}\right) = \frac{g}{g^2 + b^2}, \qquad \operatorname{Re}\left(\frac{1}{Y_{11}}\right) \neq \frac{1}{\operatorname{Re}(Y_{11})}. \tag{36}$$

Likewise, $|Z|$ includes reactance and is not resistance. Keep full complex precision throughout the transformations.

For completeness, the equal-real-reference matrix conversions are

$$\mathbf{Z} = Z_0(\mathbf{I} + \mathbf{S})(\mathbf{I} - \mathbf{S})^{-1}, \qquad \mathbf{Y} = \frac{1}{Z_0}(\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1}. \tag{37}$$

These parameter representations can be singular for legitimate ideal circuits. A series-only two-port has singular $\mathbf{Y}$ but a perfectly meaningful series impedance. Use a representation suited to the identified network.

8 When the result can be compared with Q3D

The comparison must use the same current path

Q3D resistance matrices and a two-port line resistance are not automatically the same observable. Q3D distinguishes partial resistance/inductance matrices and quantities formed through return-path reduction [8]. Record the complete Q3D trace expression, matrix name and reduction, then determine which combination corresponds to the SIwave path.

In a simplified two-branch matrix description, reversing the return current gives

$$\mathbf{i} = \begin{bmatrix} I \\ -I \end{bmatrix}, \qquad R_{\text{loop}} = R_{ss} + R_{rr} - R_{sr} - R_{rs}. \tag{38}$$

This formula applies only when the matrix branches and polarities match that description. Use the actual Q3D terminal definitions rather than assuming this matrix structure for an unseen model.

What makes the distributed extraction defensible

1. The ports are at the ends of the same conductor-and-return structure, with documented polarities and reference conductors.
2. One line mode dominates, and the chosen section is uniform enough for the RLGC model. Multiconductor structures generally require more ports and matrix extraction.
3. Launches, pads, connectors and discontinuities are either excluded, modeled separately or de-embedded. Significant lumped loads are not absorbed into an unexplained conductor resistance.
4. The solver models finite conductivity and the intended geometry/material effects. Loss cannot be recovered if it was absent from the exported model.
5. Geometry, return path, conductivity, temperature and material assumptions match the Q3D comparison. Frequency and DC-to-AC treatment are also aligned.

SIwave SYZ data are terminal network data. The extraction equations do not depend on the exporting tool's brand; their applicability depends on the model represented by the export. Q3D's 2024 R2 AC RL settings describe a quasi-magnetostatic solution and a DC-to-AC transition option [9]. Agreement at 1 GHz must be justified by electrical size and model assumptions, not by frequency alone.

When a single S2P cannot answer the question

The simplest non-uniqueness example is two internal series resistors:

$$z_s = R_a + R_b + j\omega L. \tag{39}$$

Changing $R_a \to R_a + \Delta$ and $R_b \to R_b - \Delta$ leaves all terminal S-parameters unchanged while changing the loss assigned to each resistor. Thus no algorithm can distinguish those internal resistances from the same S2P alone.

For an arbitrary branched board, extracting a particular trace's copper ACR needs additional information: an isolated path model, more terminal data, a justified circuit decomposition or conductor-specific field-loss data. Numerical agreement with Q3D alone is not enough to validate the physical interpretation.

9 Numerical validation on a known passive line

This test validates the *distributed extraction*, not merely an S-to-Z round trip. Construct a uniform line of length $\ell = 0.05\,\mathrm{m}$ with known integrated series impedance

$$q(s) = r_0 + sL_0 + \frac{r_1 sL_1}{r_1 + sL_1}, \qquad s = j\omega, \tag{40}$$

where $r_0 = 0.01\,\Omega$, $r_1 = 0.09\,\Omega$, $L_0 = 1\,\mathrm{nH}$ and $L_1 = 10\,\mathrm{nH}$. This is a passive resistor–inductor impedance: a series resistor, a series inductor and a resistor in parallel with an inductor. Set $z = q/\ell$, $y = sC'$, $C' = 100\,\mathrm{pF/m}$ and $G' = 0$.

The independently known total resistance is

$$\boxed{R_{\mathrm{known}}(\omega) = r_0 + \frac{r_1(\omega L_1)^2}{r_1^2 + (\omega L_1)^2}.} \tag{41}$$

Generate $\gamma, Z_c, \mathbf{T}$ from this model, then generate S using

$$\Delta = A + B/Z_0 + CZ_0 + D, \tag{42}$$

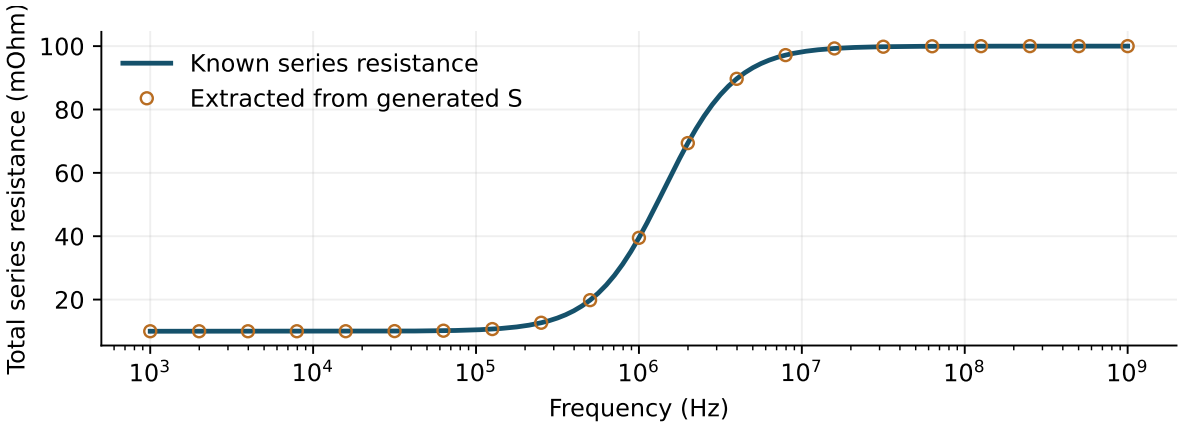
$$S_{11} = (A + B/Z_0 - CZ_0 - D)/\Delta, \qquad S_{21} = 2/\Delta, \tag{43}$$

$$S_{12} = 2(AD - BC)/\Delta, \qquad S_{22} = (-A + B/Z_0 - CZ_0 + D)/\Delta. \tag{44}$$

Use $Z_0 = 50\,\Omega$. Convert these complex S values back to ABCD and calculate $R_{\mathrm{extracted}} = \mathrm{Re}\{F(\theta)B\}$.

Frequency	Known R (mΩ)	Extracted R (mΩ)	Relative error
1 kHz	10.000044	10.000044	5.1×10^{-14}
10 kHz	10.004386	10.004386	8.0×10^{-14}
100 kHz	10.436522	10.436522	4.1×10^{-13}
1 MHz	39.491237	39.491237	7.3×10^{-14}
10 MHz	98.190547	98.190547	2.4×10^{-15}
100 MHz	99.981538	99.981538	3.4×10^{-14}
1 GHz	99.999815	99.999815	5.1×10^{-14}

The maximum relative resistance error over 241 logarithmically spaced points from 1 kHz to 1 GHz was 1.5×10^{-12}. The recovered shunt conductance was numerically zero (maximum absolute error $6.3 \times 10^{-18}\,\mathrm{S}$ for $\ell G\prime$).



Interpretation. The known and recovered curves agree to floating-point precision. The line is short enough over this test band to avoid phase-branch wrapping. The chosen frequency dependence is a realizable test model, not a claim to reproduce a copper skin-effect law. No SIwave or Q3D electromagnetic solve was run. DC follows the 10 mΩ limit and is not included in the nonzero-frequency inverse test.

10 Processing an actual SIwave export

Step 1: parse the exported complex data correctly

Read the Touchstone option line and any version-specific keywords [7]. For example, `# GHz S MA R 50` specifies GHz, scattering parameters, magnitude/angle and a 50-Ω reference. The filename alone does not determine the data type.

$$S_{ij} = a + jb \qquad \text{for RI,} \tag{45}$$

$$S_{ij} = m\exp(j\phi\pi/180) \qquad \text{for MA,} \tag{46}$$

$$S_{ij} = 10^{d/20}\exp(j\phi\pi/180) \qquad \text{for DB.} \tag{47}$$

Legacy two-port order is $S_{11}, S_{21}, S_{12}, S_{22}$, not ordinary row-major order. Touchstone 2.x can specify an alternate order and per-port references. Preserve comments containing terminal information; convert frequency to Hz and all resistance comparisons to ohms.

Step 2: test the proposed model

For a reciprocal uniform line, useful checks are

$$S_{12} \approx S_{21}, \qquad A \approx D, \qquad AD - BC \approx 1. \tag{48}$$

They are necessary diagnostics, not sufficient evidence of uniformity. A symmetric lumped network can pass them. Combine them with geometry and port information. For equal-real-reference passive terminal data, $\mathbf{I} - \mathbf{S}^H\mathbf{S}$ should be positive semidefinite within numerical tolerance.

Step 3: extract and reconstruct

De-embed known launch networks before extracting the line. With consistent chain conventions,

$$\mathbf{T}_{\text{meas}} = \mathbf{T}_L\mathbf{T}_{\text{line}}\mathbf{T}_R, \qquad \mathbf{T}_{\text{line}} = \mathbf{T}_L^{-1}\mathbf{T}_{\text{meas}}\mathbf{T}_R^{-1}. \tag{49}$$

Do not invent the launch matrices. Their extraction requires an appropriate calibration/reference structure. Choose and track θ, then calculate all four RLGC terms. Regenerate S from the extracted model and compare complex values at every frequency. A small reconstruction error verifies consistency with the model; it does not prove internal loss partitioning or resolve every phase-branch ambiguity.

Step 4: compare with Q3D at common frequencies

Compare the physically corresponding path quantity, using

$$\eta(f) = \frac{R_{\text{extracted}}(f)}{R_{\text{Q3D}}(f)}, \qquad \varepsilon(f) = \frac{|R_{\text{extracted}} - R_{\text{Q3D}}|}{|R_{\text{Q3D}}|}. \tag{50}$$

State interpolation if grids differ and avoid a relative-error metric where the denominator approaches zero. For the reported $\eta = 0.005$–0.01, inspect units, scale factors, terminal quantities and model fit before changing solver settings. A nearly constant ratio is a clue, not a diagnosis. A frequency-dependent discrepancy can also involve capacitive bypass, phase branches, transitions or distributed effects.

11 Precision, validation and an implementation algorithm

Why very small resistance is sensitive

For a small series impedance, $S_{21} = 2/(2 + z_s/Z_0)$ and

$$1 - S_{21} \approx \frac{z_s}{2Z_0}. \tag{51}$$

At $Z_0 = 50\,\Omega$, a purely resistive $1\,\mathrm{m}\Omega$ gives $1 - S_{21} \approx 10^{-5}$. Differentiating the series-through inverse gives

$$\delta z_s \approx -\frac{2Z_0}{S_{21}^2}\delta S_{21}. \tag{52}$$

Near unity transmission, an absolute complex error of 10^{-6} can correspond to approximately $0.1\,\mathrm{m}\Omega$ of impedance error. High export precision is necessary, but it cannot repair inaccurate solver data or a wrong physical model. A tiny real part can be inaccurate even when the complex impedance looks stable.

Algorithm suitable for a scripted postprocessor

1. Read frequency, reference impedances, port order and complex S. Reject unsupported normalization instead of guessing.
2. Obtain line-only data using an independently justified de-embedding model if needed.
3. Compute ABCD and record reciprocity, symmetry and passivity diagnostics.
4. Select a physically justified initial θ. Across frequency, enumerate relevant $\pm\operatorname{acosh}(x) + j2\pi k$ candidates and track the branch using continuity and delay information.
5. Evaluate $F(\theta)$ with a small-argument series when appropriate. Flag vanishing transmission and singular propagation points.
6. Compute $\ell z = FB$ and $\ell y = FC$. Export total R in ohms and R' in ohms/metre as distinctly named columns.
7. Reconstruct S, compare against the original complex data, test sensitivity to precision, and compare only the corresponding Q3D path.

Negative loss terms, abrupt discontinuities or large reconstruction errors require investigation. Do not clip them to zero to make a plot look plausible. Frequency-by-frequency passivity checks do not establish broadband causality; a fitted physical model may need an additional causality assessment.

Reusable LaTeX syntax

```
\theta(f)=\operatorname{acosh}\!\left(\frac{A(f)+D(f)}{2}\right)
R_{\mathrm{AC}}(f)=\operatorname{Re}\!\left[
  \frac{\theta(f)B(f)}{\sinh\theta(f)}\right]
R'(f)=\frac{R_{\mathrm{AC}}(f)}{\ell}
```

The core complex calculations are scriptable with IronPython. The practical work is robust parsing, branch handling and matching the physical quantity. No Ansys-specific API is needed merely to postprocess a file. This guide validates mathematics using CPython; it does not claim execution inside SIwave's IronPython environment.

12 Research papers and supporting documentation

Suggested reading order: start with the open-access derivation in [1], then [2] for resistance/de-embedding concerns, and [3] for a power-distribution example. Reference [4] is the classic interconnect paper.

[1] **Fang, W.; Ren, Y.; Liu, X.; Jiao, X.; Chu, C. (2018).** "Parameter extraction method for the twisted pair cable with rectangular connectors." *PLOS ONE* 13(10), e0205072.
DOI: 10.1371/journal.pone.0205072 Open-access full text.
Relevant content: conventional S/ABCD-to-RLGC extraction, adapter de-embedding and branch/discontinuity issues. Its experimental cable and frequency band are not a validation of the user's PCB.

[2] **Degerstrom, M. J.; Gilbert, B. K.; Daniel, E. S. (2008).** "Accurate resistance, inductance, capacitance, and conductance (RLCG) from uniform transmission line measurements." *IEEE EPEP*, pp. 77–80.
DOI: 10.1109/EPEP.2008.4675881 Author-institution record.
Relevant content: uniform-line extraction and THRU-bisect de-embedding. The accessible institutional abstract specifically notes removing resonances in distributed resistance. Full IEEE text may require institutional access.

[3] **Hasırcı, Z.; Çavdar, İ. H.; Öztürk, M. (2018).** "RLGC(f) modeling of a busbar distribution system via measured S-parameters at CENELEC and FCC bands." *Turkish Journal of Electrical Engineering and Computer Sciences* 26(1), pp. 489–500.
DOI: 10.3906/elk-1606-404 Publisher page and download.
Relevant content: frequency-dependent RLGC fitting for busbar two-port connections. Supports applicability to power-distribution structures; it does not establish universal validity up to 1 GHz.

[4] **Eisenstadt, W. R.; Eo, Y. (1992).** "S-parameter-based IC interconnect transmission line characterization." *IEEE Transactions on Components, Hybrids, and Manufacturing Technology* 15(4), pp. 483–490.
DOI: 10.1109/33.159877 IEEE publisher record.
A foundational reference also listed by MathWorks [5]. Its full IEEE text was not retrieved for this guide; the present derivation is given explicitly rather than attributed to uninspected equation numbers.

Source-access note. The PLOS full text, author-institution record for [2], publisher abstract/metadata for [3], and MathWorks documentation were accessed. Source summaries above describe only what those retrieved materials support. The mathematical derivations and the passive numerical example in this guide are worked out independently.

Supporting documentation and practical conclusion

[5] **MathWorks: `s2rlgc`.**
Convert S-parameters to RLGC transmission-line parameters.
Official implementation documentation; defines per-unit-length outputs and cites research references. Its existence confirms that S-to-RLGC is an established computational task.

[6] **Keysight E5061B: Measurement Method.**
Reflection, series-through and shunt-through impedance formulas.
Supports the distinct fixture-specific RF formulas of Section 6.

[7] **IBIS Open Forum: Touchstone File Format Specification 2.0.**
Full specification PDF.
Defines option-line interpretation, complex formats, phase units, port ordering and reference keywords. Apply the specification corresponding to the actual export version.

[8] **Ansys Q3D: Using Reduce Matrix > Return Path.**
Ansys documentation: return-path reduction.
The accessed page is from 2025 R1 and is used for the distinction between partial matrices and a modeled closed-loop quantity, not to assert a changed 2024 R2 interface.

[9] **Ansys Q3D 2024 R2: AC RL analysis settings.**
Setting AC Resistance and Inductance Analysis Parameters.
Version-specific documentation on quasi-magnetostatic AC RL and transition-region settings.

Decision for the intended task
Yes: extracting $R(f)$ from a suitable SIwave S2P is mathematically valid and can be scripted. Use the distributed RLGC derivation for a uniform path and the appropriate lumped model for a short path.
Not identifiable from a generic S2P alone: copper resistance assigned to an arbitrary individual internal conductor or a unique internal loss decomposition. Additional physical information is required.
Still unverified: the user's existing code and its 0.5–1% ratio to Q3D. A definitive audit requires that code, the original S2P, both CSVs and their units, the complete Q3D trace/reduction, and both models' terminal assignments.